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Predicting water dynamics on nanopatterned surfaces using convolutional neural network algorithm

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Yi Xiao,[1] Xulong Fan,[1] and Li Zeng[1,2,a)]

AFFILIATIONS

[1]School of Physics and Electronics, Nanning Normal University, Nanning 530001, China
[2]Guangxi Key Laboratory of Functional Information Materials and Intelligent Information Processing, Nanning Normal University, Nanning 530001, China

[a)]**Author to whom correspondence should be addressed:** zengli@nnnu.edu.cn

ABSTRACT

Nanopatterns are widespread on surfaces and play a crucial role in water dynamics, which is particularly significant in the fields of surface science, electronic devices, and advanced materials. However, the relationship between nanopatterned surfaces and water diffusion is complex and requires further research. In this paper, we use a convolutional neural network (CNN) algorithm in conjunction with molecular dynamics (MD) simulation results as a database to predict the water diffusion behavior influenced by the nanopatterned distribution of oxygen-containing groups on graphene oxide (GO) surfaces. The results show that the CNN algorithm performs effectively in predicting water diffusion on regularly nanopatterned GO surfaces, achieving a Pearson correlation coefficient (r) of 0.984 and a coefficient of determination (R^2) of 0.967. Furthermore, this trained CNN model was extended to predict the diffusion of water molecules on unseen irregularly nanopatterned GO surfaces, yielding an r of 0.956 and an R^2 of 0.857, indicating promising but preliminary generalization capability. This work presents a novel methodology for predicting the water dynamics on nanopatterned surfaces, aiding in the design of micro- and nanoscale flow surfaces.

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I. INTRODUCTION

Nanopatterned surfaces are ubiquitous in both natural systems (e.g., nacre,[1] wood,[2] cicada wings,[3] and gecko toes[4]) and engineered materials (e.g., electronic devices,[5] biosensors,[6] and advanced materials[7]). Notably, these nanopatterns feature well-defined geometric patterns (e.g., pillars and grooves)[8] and are surface modified with chemical functional groups.[9] It is precise that these nanopatterns in the surface play a crucial role in regulating the fluid dynamics on the surface[10–12] and surface wettability.[13] As a typical fluid, interfacial water molecules on the surface exhibit unique properties, such as forming two-dimensional ordered hexagonal structures induced by surface charge distributions, which give rise to the phenomenon of incompletely wetting water.[14,15] Furthermore, a fundamental understanding of how surface nanostructures influence water dynamics is therefore key to bridging material design with advanced functions such as key applications, including selective separation,[16–18] antifouling,[19] and energy-efficient catalysis.[20]

In recent years, extensive research has focused on the water dynamics on nanopatterned surfaces through both experimental[21–24] and molecular dynamics (MD) simulation studies.[25–29] Among diverse nanopatterned surfaces, the graphene oxide (GO) surface serves as a representative example. With its two-dimensional nanosheet morphology and abundant oxygen-containing functional groups (epoxy, hydroxyl, and carboxyl), GO provides a versatile platform that combines tunable surface chemistry with nanoscale topography.[30] In experiments, the naturally distributed hydrophilic groups, together with nanoscale wrinkles and corrugations typically observed on GO surfaces, collectively modulate the transport dynamics of interfacial water.[23,24] Notably, the spatial distribution of functional groups on GO surfaces exerts a dual effect on interfacial water dynamics, arising from the contrasting interactions of different

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domains with water molecules. Specifically, experimental observations reveal that nonoxidized domains, with nearly frictionless surfaces, facilitate ultrafast water transport,[31] while MD simulations further show that oxygen-containing groups on oxidized domains form strong hydrogen bonds with interfacial water, thereby impeding diffusion.[32,33] The interplay of these oxidized and nonoxidized domains renders the overall water dynamics on GO surfaces highly complex and underscores the need for systematic modeling approaches. Although the MD studies have achieved substantial progress in elucidating how the oxidation degrees govern interfacial water behavior on GO,[34,35] the impact of varying spatial distributions of functional groups at a fixed oxidation degree remains insufficiently explored. Analogous studies on other nanopatterned surfaces (e.g., SiO_2) have shown that even at constant hydroxyl coverage, variations in spatial arrangement can lead to notable differences in water diffusivity up to ~10%, as reported by Schrader *et al.*[27] Further investigations on soft material surfaces demonstrate that regulating the patterned distribution of surface functional groups can reshape the local hydration environment, turning the interfacial water diffusivity to as much as four times that of the slowest water adjacent to hydrophobic domains.[28] Whether similar effects occur on GO surfaces remains an open question that calls for systematic investigation. Addressing this issue requires large-scale data analysis, as the diverse diffusion behaviors on nanopatterned surfaces reflect the inherent complexity of interfacial water dynamics. While MD simulations can capture nanoscale phenomena with atomic precision, their high computational cost hampers systematic exploration across different functional group distributions. Therefore, an efficient and unified data-driven framework is needed to bridge surface chemical heterogeneity with water diffusivity, thereby enabling the rational design of next-generation nanopatterned materials.

Machine learning (ML), particularly deep learning, has emerged as a powerful approach to unravel the intricate relationships between surface structure and water dynamics.[36–40] Li *et al.* employed deep learning to directly predict the spatial distribution and diffusion dynamics of water molecules within a Nafion film on a Pt/C catalyst.[39] Similarly, Guo *et al.* integrated a customized deep learning network and molecular dynamics simulations with a self-constructed multi-attribute point cloud dataset to predict both static (adsorption density distribution) and dynamic (flow behavior) water transport properties on anisotropic Pt/graphene surfaces.[40] These studies established data-driven frameworks linking surface characteristics to water transport behaviors. To facilitate model training, both works encoded complex atomic configurations or water-related properties such as density and velocity into structured point-cloud representations for neural network learning. However, such encoding schemes often struggle to fully preserve the topological continuity and spatial heterogeneity of materials. Conventional fully connected neural networks, when applied to high-dimensional spatial data, also demand an enormous number of parameters, leading to prohibitive training costs.[41] Convolutional neural networks (CNNs) are inherently well-suited to address these challenges because they can autonomously extract hierarchical spatial features from raw structural data.[42] This advantage has made CNNs particularly effective for analyzing materials' microstructures and heterogeneous systems, as demonstrated in predicting the effective diffusivity of porous media directly from structural images[43,44] and modeling various transport phenomena within heterogeneous materials.[45,46] For nanopatterned surfaces (e.g., the GO surfaces in this work), where oxidation degrees and functional group distributions jointly dictate water behavior, the CNN's feature extraction capability becomes especially valuable. However, most existing studies on water dynamics prediction typically require additional auxiliary inputs (e.g., water state or properties) alongside nanopatterned surfaces structure information. This raises a question: Is there a more straightforward approach to predict water dynamics solely based on nanopatterned surface structure?

To address this issue, we developed a CNN-based deep learning framework that integrates GO nanopattern images with MD-derived diffusion data to capture the complex coupling between surface chemistry and water dynamics. Specifically, we first trained the CNN-based model on regularly patterned surfaces to predict diffusion properties. We then verified whether the trained CNN model can generalize to untrained, irregular configurations, to establish a generalizable framework for predicting diffusion coefficients directly from solid surface images. Beyond prediction, we analyzed interfacial water density distributions to elucidate how nanoscale chemical patterning modulates transport behavior. Ultimately, this framework is designed to establish a direct "pattern-transport" relationship, offering a versatile route for the rational design of nanostructured surfaces in applications such as desalination membranes, humidity sensors, and nanofluidic systems.

II. MATERIALS AND METHODS

A. Molecular dynamics simulation

As illustrated in Fig. 1(a), the MD simulation framework comprises a GO nanosheet and water molecules. The size of the simulation box is $5.2 \times 5.2 \times 2.7\ nm^3$, and that of the GO nanosheet is $5.165 \times 5.112\ nm^2$. To simplify, the GO model in this work is only modified by hydroxyl (-OH) functionalization. We investigated regularly nanopatterned and irregularly nanopatterned GO surfaces, as shown in Figs. 1(b) and 1(c), respectively. In Fig. 1(b), which illustrates regularly nanopatterned GO surfaces, nine localized sites arranged in a 3×3 grid were selected, each occupying an approximate area of $1.107 \times 0.943\ nm^2$. Within these sites, the carbon atoms on GO were either modified with hydroxyl groups or left unmodified, resulting in 511 distinct nanopatterned surfaces, which corresponded to the GO nanosheets with oxygen content ranging from 2.5% to 22.5% at intervals of 2.5%. Here, oxygen content is defined as the ratio of hydroxyl groups to carbon atoms in GO nanosheets.[47] In contrast, Fig. 1(c) shows irregularly nanopatterned surfaces, where both the size and position of the modified sites were no longer fixed, and the functional groups were arranged randomly without any apparent periodicity on the surface.

All MD simulations were performed under the NVT ensemble using GROMACS 2021.5 software.[48] The force field parameters of the GO were adopted from the literature.[49] Water molecules were modeled using the SPC/E water model. The long-range electrostatic interactions were computed using the Particle-Mesh Ewald[50] method with a cutoff of 1 nm. The van der Waals (vdW)[51] interactions were evaluated with a cutoff of 1 nm. Periodic boundary conditions were implemented in all directions. All simulations were

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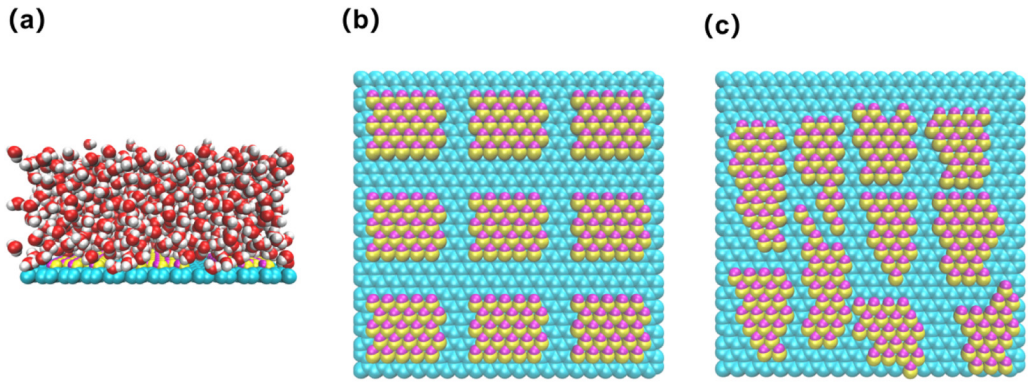


FIG. 1. (a) Snapshot of the molecular dynamics (MD) simulation system, which consisted of a GO nanosheet and water molecules. Carbon atoms in the GO nanosheet are shown in cyan, while oxygen and hydrogen atoms in the hydroxyl groups are represented as yellow and magenta spheres, respectively. For water molecules, oxygen atoms are colored red, and hydrogen atoms are colored white. (b) An example of a regularly nanopatterned surface with nine hydroxyl group-modified regions. This pattern is one representative pattern from the 511 GO models in this study. (c) An example of an irregularly nanopatterned surface, where the size and position of the modified regions are assigned randomly.

performed at a constant pressure of 1 bar and a constant temperature of 300 K. The total simulation time for each MD simulation was 12 ns. The initial 10 ns simulation was performed with a 1 fs time step to ensure that the system reached equilibrium. The final 2 ns in a 1 fs time step were utilized to collect data for analysis.

B. CNN model construction

1. Preparation of the dataset and architecture of the network model

During dataset preparation, the dataset for training the CNN model includes 511 images of GO nanosheets (as inputs) and their corresponding effective water diffusion coefficients (as outputs), which were derived from MD simulations. Images were rendered in VMD[52] with consistent visualization settings (Quicksurf modstyle, Diffuse modmaterial, carbon in violet, oxygen and hydrogen in yellow) and were resized to 250×250 pixels, as shown in Fig. 2.

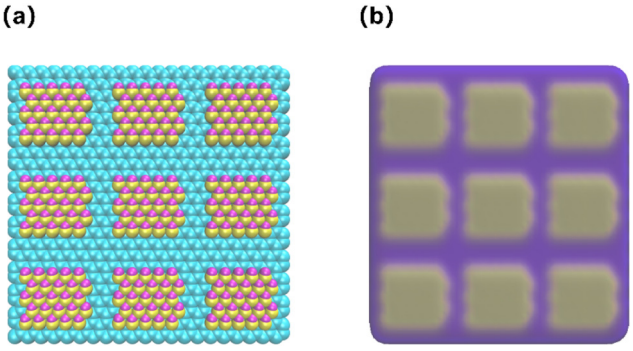


FIG. 2. (a) Example of a GO nanopatterned surface and (b) its corresponding VMD-processed visualization.

The dataset is illustrated in Fig. 3(a), where one representative image is shown for each oxidation level for clarity.

Subsequently, in the model architecture step, the CNN architecture was designed to hierarchically extract spatial features from input images and map them to a continuous diffusion coefficient value, as illustrated in Fig. 3(b). The CNN consisted of three convolutional stages followed by fully connected layers. In the first convolutional stage, two convolutional layers with 3×3 kernels and tanh activation functions were applied, each followed by batch normalization and a 2×2 max-pooling operation for downsampling. In the second and third convolutional stages, only one convolutional layer with the same configuration was used before batch normalization and max-pooling. The number of filters was kept at 64 in the first and second convolutional stages and subsequently increased to 128 in the third convolutional stage for the capture of more complex and higher-level feature representations. The extract features were then flattened and processed through two fully connected layers, each with 32 units and tanh activation functions. At last, a linear output layer predicted the water diffusion coefficient.

2. CNN network training and testing process

The CNN model employed in this study was implemented using TensorFlow[53] (version 2.13.1) in a Python 3.8 environment, with supporting libraries including Keras (for building and training deep learning models) and Matplotlib (for visualization of training progress). The hyperparameters of the Convolutional Neural Network (CNN) model are listed in Table S1 of the supplementary material. We used the Adam optimizer (learning rate = 0.001) and an L2 regularization term ($\lambda = 0.01$) that was applied to all layers to mitigate overfitting. To ensure reliable evaluation, the dataset was randomly divided into training and test sets in an 8:2 ratio. The batch size was fixed at 8, and the number of epochs was set to 1500. To evaluate and compare the water diffusion coefficients predicted by the trained CNN model with the MD simulation results,

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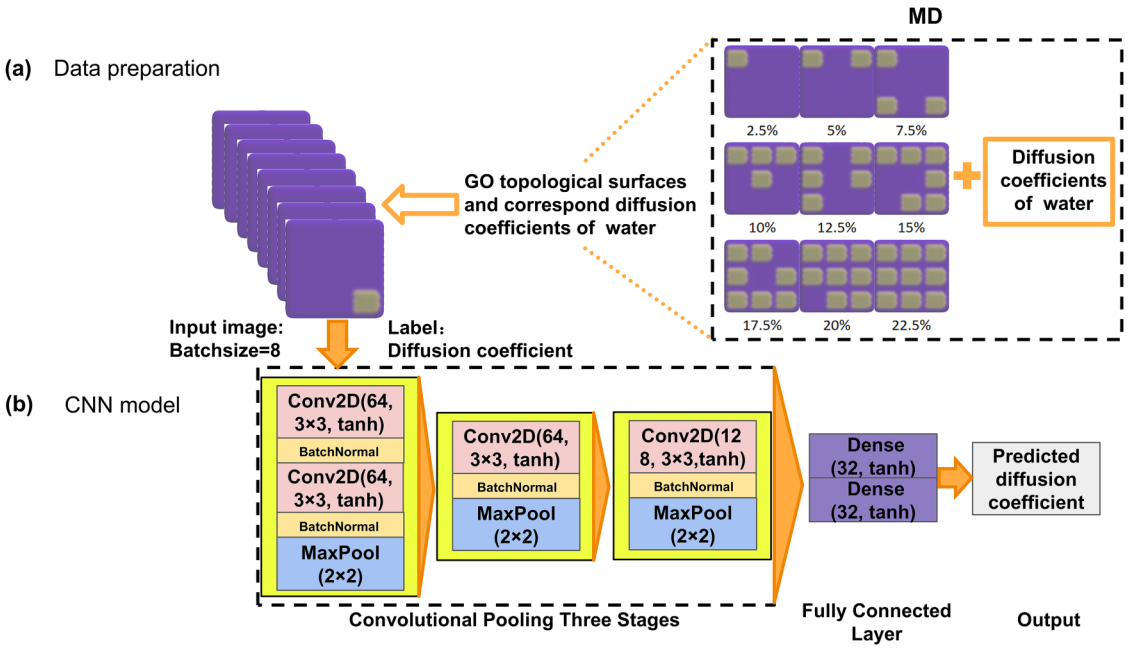


FIG. 3. (a) GO with surface oxidation ranging from 2.5% to 22.5%, visualized using the VMD software. One randomly selected image is shown for each degree of oxidation. Carbon atoms are shown in violet, and hydrogen and oxygen atoms in yellow. (b) Visualizing a simple model of CNN architecture. Conv2D (k, s): 2D convolutional layer, where k denotes the number of filters and s represents the kernel size. MaxPool (p): Pooling layer, where p indicates the size of the pooling window. Dense (n): Fully connected layer, where n denotes the number of neurons.

the accuracy of the CNN model was assessed using three commonly employed regression loss functions: Huber Loss, Mean Absolute Error (MAE), and Mean Squared Error (MSE). These loss functions were consistently applied during both training and testing to provide a quantitative measure of prediction performance.[54] The mathematical expressions of the three loss functions are given as follows:[55,56]

$$L_\delta = \begin{cases} \frac{1}{2}(y_i - \hat{y}_i)^2 & \text{if} |(y_i - \hat{y}_i)| < \delta, \\ \delta\left((y_i - \hat{y}_i) - \frac{1}{2}\delta\right) & \text{otherwise,} \end{cases} \quad (1)$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N} |y_i - \hat{y}_i|, \quad (2)$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N} (y_i - \hat{y}_i)^2. \quad (3)$$

Here, y_i represents the diffusion coefficient value obtained by MD, $\hat{y}_i$ represents the predicted diffusion coefficient value predicted by CNN, and δ is a hyperparameter.

After confirming the evaluation method, the practical prediction process of the CNN model for water diffusion coefficients was designed as follows: During the training process, the GO nanosheet images were fed to the CNN network, while the water diffusion coefficients calculated from the MD simulation corresponding to each input case were used as the label. During the testing process, the water diffusion coefficients could be predicted by inputting new GO nanosheet images in the test set into the trained networks.

3. Prediction performance evaluation

The prediction performance of the trained CNN model was evaluated using the coefficient of determination (R^2) and the Pearson correlation coefficient (r). R^2 is a widely used metric for evaluating the goodness of fit in linear regression, and r measures the correlation between two variables.[57] In this study, R^2 was computed using the r2_score function from sklearn.metrics, and r was calculated using the pearsonr function from scipy.stats, both of which are provided by widely used scientific computing libraries in Python.

III. RESULTS AND DISCUSSION

A. Investigation of regularly nanopatterned surfaces

1. MD simulation of water diffusion on regularly nanopatterned surfaces

The water diffusion coefficients D used to construct the CNN database were obtained from MD simulations. Here, the diffusion

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coefficient D of the water layer distributed from 0 to 1.35 nm in the xy-plane was calculated using the following Einstein relation:[58]

$$D = \frac{\langle |r(t) - r(0)|^2 \rangle}{4t}. \quad (4)$$

Figure 4 shows the result of the diffusion coefficients obtained from the MD simulations under different oxidation degrees of the GO surface. The diffusion coefficients range from 1.507×10^{-5} to 2.698×10^{-5} cm^2/s, consistent in order of magnitude with those reported in related studies.[59,60] Furthermore, an inverse relationship is observed between the diffusion coefficient and the oxidation degree of the GO nanosheets, that is, with the increase in oxidation degree from 2.5% to 22.5%, the diffusion coefficients decrease from 2.698×10^{-5} to 1.507×10^{-5} cm^2/s with a total difference of 1.191×10^{-5} cm^2/s. It is demonstrated that the oxidation degree of the GO nanosheets significantly affects water diffusion. In addition to the oxidation degree, it was important to note that GO surface patterning also contributes significantly to water molecular diffusion at a fixed oxidation degree. For example, the 10% oxidation degree situation, which is selected as a representative case, because it represents the highest combinatorial variety and accounts for a substantial fraction of the total 511 GO surface patterns generated, corresponding to four oxidized regions distributed across the nine binding sites that yield 126 distinct surface configurations. At this oxidation level, the diffusion coefficients varied from 1.922×10^{-5} to 2.244×10^{-5} cm^2/s with a variation of 0.322×10^{-5} cm^2/s, implying that water diffusion can also be substantially influenced by the spatial distribution of functional groups and nanoscale patterning, although the oxidation degree is a key factor. This finding is consistent with previous studies,[27] which highlight the important role of surface patterning in modulating water diffusion on GO surfaces. These results suggest the necessity for a more comprehensive analysis of surface heterogeneity and its impact on water transport.

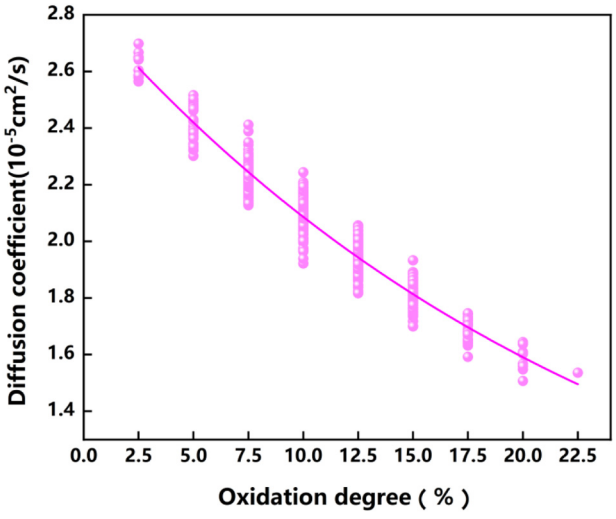


FIG. 4. Diffusion coefficient of the surface water layer with the oxidation degree of GO.

2. Prediction and evaluation of the CNN model for water diffusion

To train the CNN model for predicting water diffusion coefficient, 511 images of regularly patterned GO surface along with their corresponding diffusion coefficients were used as the original CNN dataset. This dataset was randomly partitioned into a training set (80%) and a testing set (20%) to ensure both sets were drawn from the same data distribution. The training set was fed into the CNN for model training, while the testing set was used to evaluate the accuracy of the network's predictions. During the model training process, the loss functions, including Huber Loss, MAE, and MSE, all decreased steadily with epochs, reflecting continuous performance improvement. As shown in Fig. S1 of the supplementary material, this trend indicates that the model was trained effectively without signs of overfitting. The evaluation on the test set yielded a loss of 0.002, with MAE and MSE values of 0.031 and 0.002, respectively, indicating that the prediction errors remain within a small range and demonstrate the high accuracy and robustness of the trained model. The predicted results are shown in Fig. 5, and the CNN-diffusion coefficients display a strong linear relationship with the MD-diffusion coefficients. The trained CNN model achieved a high Pearson correlation coefficient ($r = 0.984$) and coefficient of determination ($R^2 = 0.967$), both of which demonstrated the model's excellent predictive performance. In addition, we performed fivefold cross-validation and image robustness tests, with detailed results provided in Tables S2 and S3, respectively, of the supplementary material. These results further confirm the model's stability and consistency across different data partitions and image variations, demonstrating that directly predicting water diffusion coefficients with a

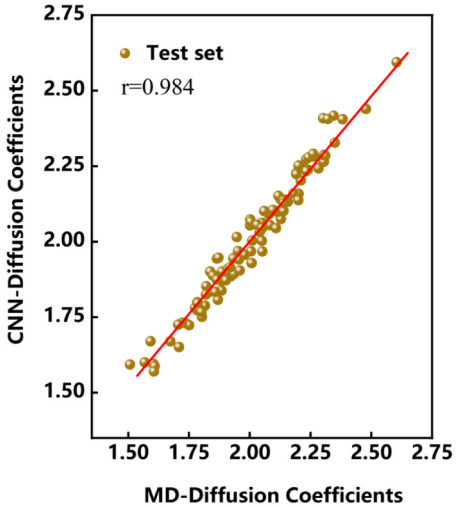


FIG. 5. Correlation plots of water diffusion coefficients calculated by MD (horizontal direction) with the predicted diffusion coefficients by CNN (vertical direction) for datasets.

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CNN, relying exclusively on nanopatterned surface structures, is both reliable and accurate.

B. Investigation of irregularly nanopatterned surfaces

1. MD simulation of water diffusion on irregularly nanopatterned surfaces

In the previous study, the CNN model was trained for regularly nanopatterned surfaces. However, in practical systems, irregular nanopatterns of oxidized domains are more common. To evaluate whether the proposed CNN model retains its effectiveness under irregularly nanopatterned surfaces and to further assess its generalization capability, an additional dataset comprising irregularly nanopatterned surfaces was constructed. This dataset included the same nine oxidation degree values (2.5% to 22.5%) as the regular nanopattern set, with each oxidation degree applied to six irregular surfaces where hydroxyl groups randomly modified carbon atoms of the GO surface, resulting in a total of 54 irregular nanopatterns for validation (as shown in Fig. S2 of the supplementary material). Notably, in contrast to the regular nanopatterns, where oxidation is restricted to fixed rectangular areas at specific sites, the irregular nanopatterns permit non-rectangular shapes and arbitrary spatial arrangements of oxidized domains, thereby better capturing the diversity and morphological randomness observed in practical systems. In Fig. 6, the shaded boxes denote the diffusion coefficient range for regular nanopatterns, while each blue point represents the diffusion coefficient on irregularly nanopatterned surfaces. It can be observed that some diffusion coefficients on irregularly nanopatterned surfaces exceed the range observed for regularly nanopatterned surfaces, exhibiting a numerical range not observed in the latter. For instance, at oxidation degrees of 17.5%, three out of six irregular nanopatterns exceeded the ranges of the regular ones, corresponding to the diffusion coefficient range from 1.794×10^{-5} to 2.001×10^{-5} cm^2/s with an interval of 0.207×10^{-5} cm^2/s, whereas the regular nanopatterns exhibit a diffusion coefficient spanning 0.154×10^{-5} cm^2/s (from 1.592×10^{-5} to 1.746×10^{-5} cm^2/s). This result is derived from the irregular distribution of nanoscale surface patterns, which introduces greater topological diversity while simultaneously affecting water diffusion behavior. Furthermore, this finding also reinforces that surface nanopattern is a significant factor that affects water diffusion behavior, even at identical oxidation degrees of the surface.

2. Prediction and application of the CNN model for water diffusion

Next, we used these 54 irregularly nanopatterned surfaces to evaluate the performance of the previously trained CNN model on such surfaces. The results shown in Fig. 7 illustrate the correlation between the water diffusion coefficients obtained from the MD simulations and those predicted by the CNN model for irregularly nanopatterned surfaces. Our analysis yielded a Pearson correlation coefficient r of 0.956 and a coefficient of determination R^2 of 0.857. Compared with the regular nanopatterned images, the irregular ones lead to a lower R^2 mainly because of their higher structural variability. Although both image types underwent identical preprocessing, the regular surfaces possess consistent rectangular features that facilitate stable feature extraction. In contrast, the irregular surfaces lack such geometric regularity, increasing pattern diversity and reducing prediction accuracy. Nevertheless, the model still achieves an $\mathbf{R}^2$ of 0.857, demonstrating good generalization to surface morphologies absent from the training set.

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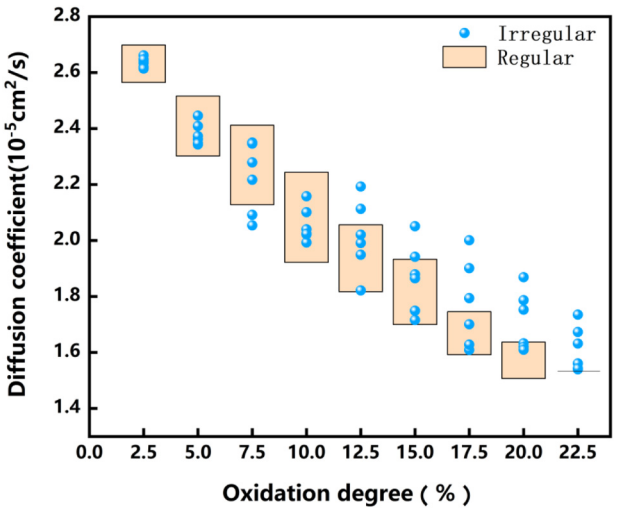


FIG. 6. A test set of 54 irregular images of GO nanopatterned surfaces, functionalized with 2.5%–22.5% hydroxyl groups. The diffusion coefficients resulting from the irregular images are shown as scattered data points, while the ranges for regular nanopatterns are represented by shaded boxes.

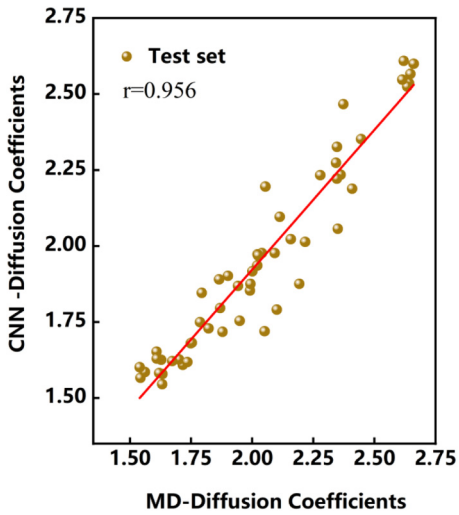


FIG. 7. Correlation plot of water diffusion coefficients calculated by MD simulation (horizontal direction) vs predicted values by CNN (vertical direction) for the 54 irregularly nanopatterned surfaces test set.

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Building upon this well-generalized model, we developed a prediction system that requires only image input to directly output water diffusion coefficients, eliminating the need for additional training or iterative optimization (see Fig. S3 of the supplementary material). In contrast to molecular dynamics simulations, which require multiple intricate steps such as constructing the nanopatterned surfaces, building the complete water-GO system, executing lengthy and computationally demanding simulations, and subsequently calculating diffusion coefficients from trajectory data, our system delivers results immediately from image input. This approach drastically reduces both time and computational resources, providing an efficient and convenient tool for rapid surface design and analysis of water dynamics. It is particularly suitable for applications such as screening functionalized surfaces for desalination membranes, humidity sensors, or advanced nanofluidic devices.

C. Surface water molecule aggregation analysis

To elucidate how the GO surface nanopatterned structure influenced water diffusion, we examined the water density profile, as water density is closely correlated with diffusion behavior.[61–63] The trajectories of water molecules within 0–1.35 nm from the surface were analyzed over a 2 ns post-equilibration period, because this region captures where basal-plane oxygen groups most strongly perturb water structure. Next, we selected oxidation degrees of 2.5%, 10%, and 22.5% as representative cases spanning low, medium, and high oxidation degrees, for both regularly and irregularly nanopatterned surfaces, resulting in six representative systems (see Fig. 8). At an oxidation degree of 2.5%, the water density peaks are primarily localized around isolated oxygen sites. This phenomenon is further confirmed at higher oxidation degrees (10% and 22.5%), and the trend remains consistent for both Fig. 8(a) regular and Fig. 8(b) irregular surface patterns. Regardless of the oxidation distribution, high-density water regions always appear in the vicinity of oxidized functional groups. This indicates that water molecules are strongly attracted by hydroxyl groups, forming distinct regions of locally enhanced density, consistent with the findings of Williams *et al.*[35] Moreover, the spatial extent and connectivity of these high-density hydration domains are inversely correlated with the water diffusion coefficient. Surfaces exhibiting sparse and isolated density regions (e.g., 2.5% oxidation) correspond to faster diffusion, whereas those forming extensive high-density water domains (e.g., 22.5% oxidation) markedly slow down diffusion. To further verify this trend, we quantified the fraction of high-density regions (with local density >1.2 g/cm^3) relative to the total surface

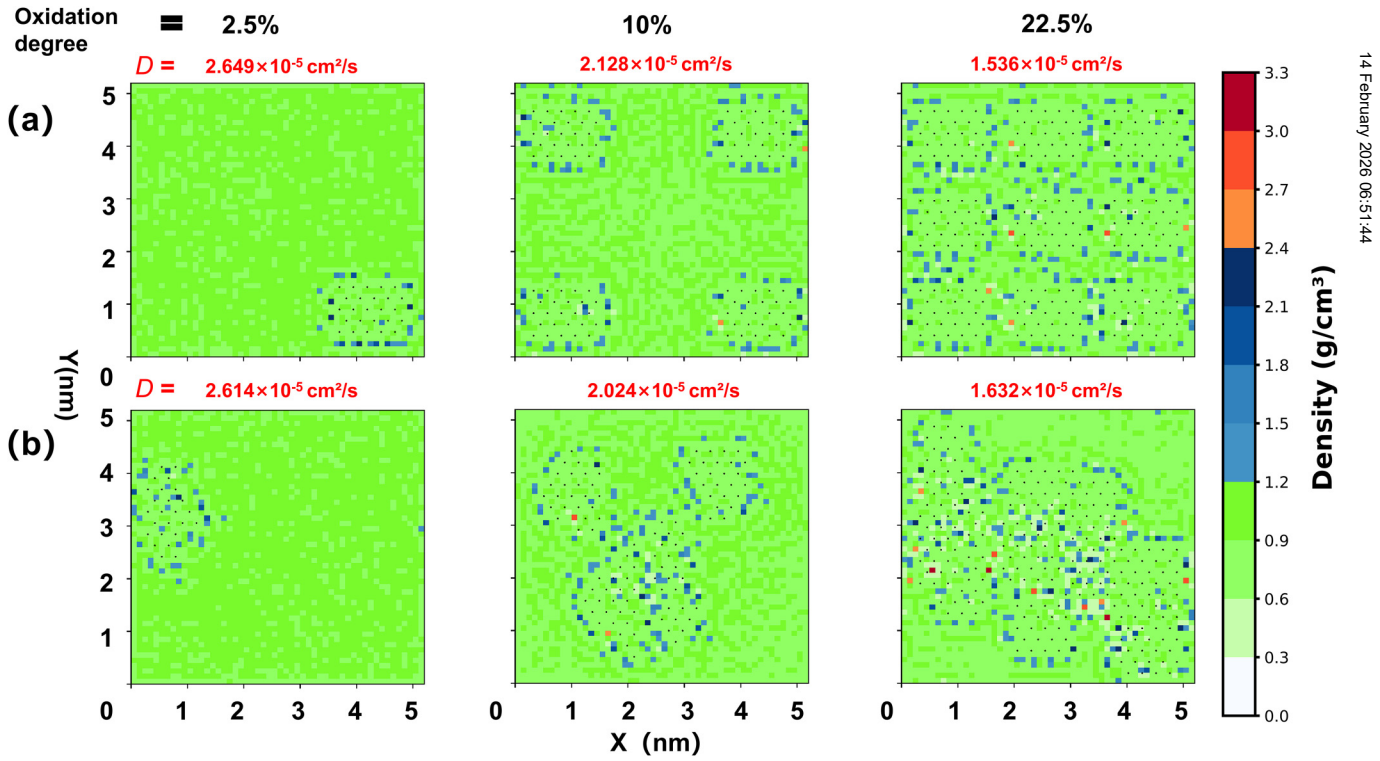


FIG. 8. Water density distributions within a 0–1.35 nm region above GO surfaces were analyzed during the 2 ns post-equilibration simulation period. The small black points denote the positions of the hydroxyl-modified carbon atoms. Results are shown for three oxidation degrees (2.5%, 10%, and 22.5%) under two surface configurations: (a) Regular and (b) irregular surface configurations, respectively.

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area. At a fixed oxidation degree, variations in the spatial distribution of surface hydroxyl groups lead to distinct surface water density patterns. As shown in Table S4 of the supplementary material, the higher the fraction of high-density regions, the lower the diffusion coefficient. This slowdown arises from the strong attraction between water molecules and surface hydroxyls, which confines water more tightly around oxidized domains and thus suppresses its diffusion, as elucidated by Devanathan *et al.*, who attributed the strong interaction to hydrogen bonding between hydroxyl groups and water.[64]

IV. CONCLUSION

In this study, we employed a CNN to predict nanoscale water dynamics on GO surfaces based solely on their nanopatterned morphology, with MD simulations serving as the reference for training. The CNN achieved high predictive accuracy and strong universality: for regularly nanopatterned surfaces, it reached an r of 0.984 and an R^2 of 0.967 between predicted and MD-calculated diffusion coefficients. Notably, when applied to untrained irregular nanopatterns, the model maintained robust performance ($r = 0.956$, $R^2 = 0.857$), underscoring its capacity to capture structure–property relationships beyond the training domain. MD simulations further revealed that water diffusion coefficients decrease with increasing oxidation degree and are modulated by the spatial arrangement of oxidized domains. Density analysis provided mechanistic insight, showing that water molecules preferentially accumulate near OH groups along nanopattern edges, resulting in heterogeneous interfacial distributions that directly influence diffusion. These findings establish a generalizable pathway linking solid surface morphology to liquid transport properties, thereby bridging nanoscale features and macroscopic behavior. This work presents an efficient computational framework for predicting water transport, and future extensions will incorporate additional factors, such as varied functional groups and applied pressure, to further advance the rational design of functional membrane materials.

SUPPLEMENTARY MATERIAL

The supplementary material includes hyperparameters of the Convolutional Neural Network (CNN) model, a numerical figure of the loss function vs the number of iterations during CNN training on regular-patterned surfaces, results of 5-fold cross-validation, data from the image robustness test, examples of regular and irregular hydroxyl functionalization patterns on GO surfaces, prediction system diagrams (including the prediction process for direct prediction on irregular-patterned surfaces), and a detailed data table on the fraction of high-density regions in the density distribution.

ACKNOWLEDGMENTS

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yi Xiao: Data curation (lead); Formal analysis (lead); Investigation (lead); Writing – original draft (lead). **Xulong Fan:** Data curation (supporting); Investigation (supporting). **Li Zeng:** Conceptualization (lead); Funding acquisition (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supplementary material for “Predicting water dynamics on Nanopatterned surfaces using Convolutional Neural Network algorithm”

Yi Xiao[1], Xulong Fan[1], Li Zeng[1,2*]

1.School of Physics and Electronics, Nanning Normal University, Nanning 530001, China

2.Guangxi Key Laboratory of Functional Information Materials and Intelligent Information Processing, Nanning Normal University, Nanning 530001, China

hyperparameter	value
learning rate	0.001
batch size	8
random horizontal flip	enabled
random vertical flip	enabled
L2 regularization (λ)	0.01

Table S1. Hyperparameters of the Convolutional Neural Network (CNN) Model

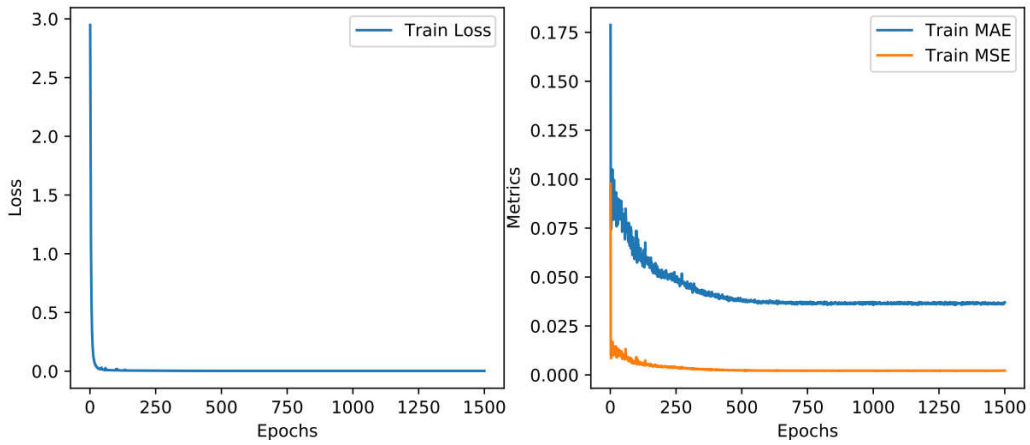


Fig. S1. The training Huber Loss, MAE, and MSE demonstrate stable convergence throughout the training process, confirming the model's good performance and generalization.

Results of 5-fold Cross-Validation (k-fold CV):

We performed an additional 5-fold cross-validation independent of the main training process. Using KFold (n_splits=5, shuffle=True, random_state=42), the entire training dataset (from train_data_dir) was randomly partitioned into 5 mutually exclusive subsets (each accounting for 20% of the total training data); in each fold the model was re-initialized, trained on the combined data of 4 subsets (totaling 80% of the training data) and validated on the remaining 1 subset (the 20% validation partition for that fold). The results of the 5-fold cross-validation are shown in the table below:

fold	best_val_MSE	best_val_MAE	val_R^2
1	0.002	0.030	0.963
2	0.001	0.029	0.976
3	0.002	0.029	0.961
4	0.002	0.035	0.969
5	0.002	0.038	0.940

Table S2. Five-Fold Cross-Validation results of the CNN Model

Data from the Image Robustness Test：

We conducted an independent robustness assessment in which the original VMD-rendered images were systematically modified across multiple dimensions, including pixel resolution (128×128 and 250×250), grayscale modes (L and LA), added image noise (Gaussian and Salt-and-pepper), and different levels of unified lighting. The detailed outcomes of these robustness tests are summarized in the table below：

presentational form	pixel	additional noise	additional light settings	R^2
Grayscale(L)	250×250	No	No	0.964
Grayscale(L)	250×250	Gaussian	No	0.963
Grayscale(LA)	250×250	No	No	0.961
Grayscale(LA)	250×250	Gaussian	No	0.962
Grayscale(L)	250×250	No	20%	0.969
Grayscale(L)	250×250	Gaussian	20%	0.964
Grayscale(LA)	250×250	No	20%	0.965
Grayscale(LA)	250×250	Gaussian	20%	0.963
Grayscale(L)	250×250	No	50%	0.964
Grayscale(L)	250×250	Salt-and-Pepper	50%	0.969
Grayscale(LA)	250×250	No	50%	0.971
Grayscale(LA)	250×250	Salt-and-Pepper	50%	0.967
Grayscale(L)	128×128	No	No	0.861
Grayscale(L)	128×128	Gaussian	No	0.620
Grayscale(LA)	128×128	No	No	0.821
Grayscale(LA)	128×128	Gaussian	No	0.612
Grayscale(L)	128×128	No	20%	0.946
Grayscale(L)	128×128	Gaussian	20%	0.945
Grayscale(LA)	128×128	No	20%	0.949
Grayscale(LA)	128×128	Gaussian	20%	0.945
Grayscale(L)	128×128	No	50%	0.950
Grayscale(L)	128×128	Salt-and-Pepper	50%	0.908
Grayscale(LA)	128×128	No	50%	0.950
Grayscale(LA)	128×128	Salt-and-Pepper	50%	0.924

Table S3.CNN prediction robustness under different rendering and Image-Processing variations

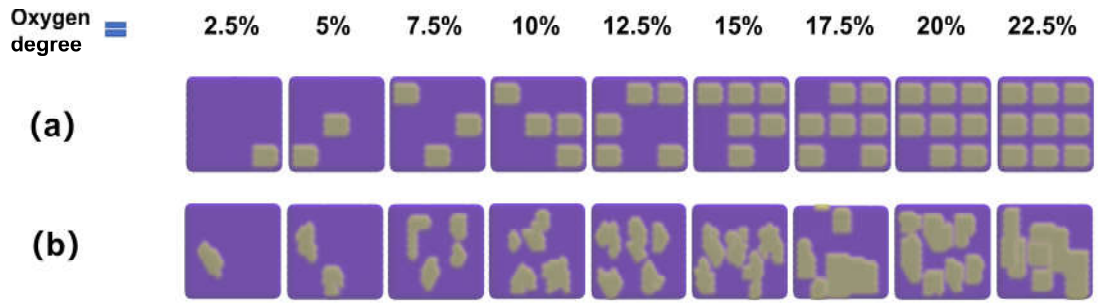


Fig. S2. Examples of (a)regular and (b)irregular hydroxyl functionalization patterns on GO surfaces

```
zl4090@zl4090:/mnt/ssd/XY$ python -u "/mnt/ssd/XY/GO/GO/Data prediction system.py"
2025-05-19 11:43:45.055105: I tensorflow/core/util/port.cc:110] oneDNN custom operations
rom different computation orders. To turn them off, set the environment variable `TF_ENAB
2025-05-19 11:43:45.056299: I tensorflow/tsl/cuda/cudart_stub.cc:28] Could not find cuda
2025-05-19 11:43:45.077223: I tensorflow/tsl/cuda/cudart_stub.cc:28] Could not find cuda
2025-05-19 11:43:45.077538: I tensorflow/core/platform/cpu_feature_guard.cc:182] This Ten
To enable the following instructions: AVX2 AVX512F AVX512_VNNI AVX512_BF16 FMA, in other
2025-05-19 11:43:45.498346: W tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38] TF-TR

=== Diffusion Coefficient Prediction System ===
1. Single image prediction
2. Batch prediction
3. Exit
2025-05-19 11:43:45.976872: I tensorflow/compiler/xla/stream_executor/cuda/cuda_gpu_execu
ast one NUMA node, so returning NUMA node zero. See more at https://github.com/torvalds/l
2025-05-19 11:43:45.992910: W tensorflow/core/common_runtime/gpu/gpu_device.cc:1960] Cann
d properly if you would like to use GPU. Follow the guide at https://www.tensorflow.org/i
Skipping registering GPU devices...

Please enter operation number: 1
Please enter image path: /mnt/ssd/XY/GO/GO/random54/1.png
Prediction result: 2.5996

Please enter operation number: 2
Please enter image directory path: /mnt/ssd/XY/GO/GO/random54
Batch prediction completed. Results saved to: /mnt/ssd/XY/GO/GO/batch_predictions.csv

Please enter operation number: 3
Thank you for using, goodbye!
```

Fig. S3. Example of the diffusion coefficient prediction system interface. The system supports single-image prediction (1, output: 2.5996) and batch prediction (2, results automatically saved as CSV), providing a user-friendly workflow for CNN-based diffusion coefficient prediction.

Oxidation degree	2.5%	10%	22.5%
(a) high-density	0.92%	3.96%	7.32%
$D_{(a)}$	2.649×10^{-5} cm²/s	2.128×10^{-5} cm²/s	1.536×10^{-5} cm²/s
(b) high-density	1.33%	4.33%	6.25%
$D_{(b)}$	2.614×10^{-5} cm²/s	2.024×10^{-5} cm²/s	1.632×10^{-5} cm²/s

Table S4. Fraction of high-density water regions (> 1.2 g/cm³) and the corresponding diffusion coefficients (D) at different oxidation degrees. (a) and (b) correspond to two independent GO surface configurations with identical oxidation degrees.